

CSU San Marcos
 Department of Mathematics
 333 Twin Oaks Valley Rd.
 San Marcos, CA 92078

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EDUCATION

Ph.D. in Mathematics, Iowa State University 2015
 Dissertation: **Graph Determined Symbolic Dynamics and Hybrid Systems**.
 Adviser: Dr. Wolfgang Kliemann

B.A. in Mathematics, Bowdoin College, Brunswick, Maine 2010
 Honors Thesis: "Stochastic Perturbations of the Fitzhugh-Nagumo Equations"
 Adviser: Dr. Mary Lou Zeeman.

EMPLOYMENT

Associate Professor, California State University San Marcos 2026 - present
Assistant Professor, California State University San Marcos 2021 - 2026
Assistant Professor, Carroll College 2019 - 2021
Visiting Assistant Professor, Pomona College 2016 - 2019

HONORS AND AWARDS

Project NExT Fellow, Peach '18 Cohort 2018-2019
 Aggie Ho Research Award, Iowa State University 2015
 Graduate College Research Excellence Award, Iowa State University 2014
 Edward S. Hammond Prize, Mathematics Department, Bowdoin College 2010
 Graduated with Honors, Mathematics Department, Bowdoin College 2010

PUBLICATIONS AND PAPERS

9. W. Aitken, K. Ayers, H. Smith. "General Boyd-Lawton theorems with multivariable limits." Submitted, 2025.
8. K. Ayers, M. Kooiker. "A structural analysis of population graphs." Submitted, 2025.
7. K. Ayers, A. Radunskaya. "Stability of the invariant distribution of the stochastic logistic map." Conditionally accepted in *Research in the Mathematics Sciences*, January 2025.
6. K. Ayers, D. Dmitrishin, A. Radunskaya, A. Stokolos. "Search for invariant sets of the generalized tent map." *Journal of Difference Equations and Applications*, 29(9-12), 1156-1183, 2022. <https://doi.org/10.1080/10236198.2022.2119851>
5. K. Ayers. "Chain Recurrence in Graph Determined Hybrid Systems". *Journal of Dynamics and Control Systems* (2019). <https://doi.org/10.1007/s10883-019-09440-x>

4. H. Cho, K. Ayers, L. de Pillis, Y-H. Kuo, J. Park, A. Radunskaya, R. Rockne. "Modeling acute myeloid leukemia in a continuum of differentiation states." *Letters in Biomathematics*, 5(sup1): S69-S98, 2018.
3. K. Ayers. "Graph Determined Symbolic Dynamics and Hybrid Systems," Ph.D. dissertations, Iowa State University: ProQuest/UMI, 2015
2. J. Ackerman, K. Ayers, E. Beltran, J. Bonet, D. Lu, T. Rudelius. " A behavioral characterization of Markov chains and discrete time dynamical systems over directed graphs." *Qualitative Theory of Dynamical Systems*, 13(1):161-180, 2014.
1. K. Ayers, X. Garcia, J. Kunze, T. Rudelius, A. Sanchez, S. Shao, E. Speranza. "Limit and Morse sets of deterministic hybrid systems." *Qualitative Theory of Dynamical Systems*, 12(2):335-360, 2013.

GRANTS AND FELLOWSHIPS

NSF Improving Undergraduate Stem Education (IUSE)	2025-2028
Title: "Collaborative Research: Elevating the calculus standard of care through an innovative collaborative testbed"	
CSUSM PI	\$173,098
AMS PUI Travel Grant	2023
Joint Mathematics Meetings	\$2100
AMS PUI Travel Grant	2022
Joint Mathematics Meetings - on hold	\$2,100
NIH Research Enhancement Award (R15)	2021-2024
Title: "The Architecture of Missing and Archaic Variation in Human Population Genomic Data"	
PI	\$440,087
AIM Math Community	2020-present
"Little School Dynamics - A Dynamical Systems Community Projects for Researchers at PUIs"	
Co-PI	

PROFESSIONAL PRESENTATIONS

"The Stochastic Logistic Map: Theory and Applications"	2026
Invited Colloquium Talk, Claremont Center for the Mathematical Sciences	

“Graph-Theoretic Models of Genetic Relationships” Invited Paper Session, MathFest 2026	2026
“Evolutionary Dynamics, Population Graphs, and Ghost Populations” International Society for Mathematical Biology Conference, Seoul, Korea	2024
“Stability of the invariant distribution of the stochastic logistic map” JMM AIM Special Session on Little School Dynamics: Cool Research Done at PUIs	2024
“Stability of the invariant distribution of the stochastic logistic map” Invited Talk, University of Utah Applied Mathematics Seminar	2024
“Stability of the invariant distribution of the stochastic logistic map” CSU Mathematical Sciences Conference	2023
“A Chaotic Introduction” Route 78 Math Day	2023
“Search for invariant sets of the generalized tent map” JMM AIM Special Session on Little School Dynamics: Cool Research Done at PUIs	2023
“An Exploration of Sharkovskii’s Theorem” University of San Diego Invited Colloquium Talk	2022
“Baby Sharkovskii” San Marcos Informal Mathematics In-Person Colloquium	2022
“Selected Topics in Hybrid Systems” Invited Talk, Boston Computing Group	2022
“Random Perturbations of Self-Regulating Biological Processes” JMM AWM Special Session on Women in Mathematical Biology	2022
“Harris Irreducibility of the Stochastic Logistic Map” JMM AMS Special Session on Little School Dynamics: Cool Research done at PUIs	2022
“Fixed Points, Stability, and Oscillations: How can we manipulate systems?” Invited Colloquium Talk, Cal State Long Beach	2022
“Fixed Points, Stability, and Oscillations: How can we manipulate systems?” Invited Colloquium Talk, Skidmore College	2022
“Little School Dynamics - Research at PUIs” Invited Talk, SIMIODE Expo	2022
“Random Perturbations of Self-Regulating Biological Processes” UCSD NITMB Colloquium	2021
“Fixed Points, Stability, and Oscillations: How can we manipulate systems?” San Marcos Mathematics Online Colloquium	2021
“Stochastic Logistic Maps and Invariant Distributions” Invited Graduate Seminar, University of Georgia	2021

“Stochastic Perturbations of Unimodal Maps” Invited Colloquium Talk, Cal State San Marcos	2021
“Perceiving Patterns and Chaos” Cal State San Marcos Student Seminar	2021
“Random Perturbations of Unimodal Maps” AIM Math Community - Little School Dynamics Invited Talk	2021
“Faculty Dynamics Research Opportunities - Primarily Undergraduate Institutions” Invited Panel, SIMIODE Expo	2021
“Stabilization of Quasi-Periodic Orbits of the Random Logistic Map” Invited Talk, Women in Math in Southern California Conference	2020
“Chain Recurrence in Graph Determined Hybrid Systems” MAA Contributed Paper Session, Joint Mathematics Meetings 2020	2020
“A Skew Product Model for Hybrid Dynamical Systems” Invited Colloquium Talk, University of Montana	2019
“Stochastic Perturbations of the Logistic Map.” MAA Invited Paper Session, MathFest 2019	2019
“Perceiving Patterns and Chaos” Invited Colloquium Talk, Carroll College	2019
“A Skew Product Model for Hybrid Dynamical Systems” Invited Colloquium Talk, California Polytechnic Institute, Pomona	2018
“A Chaotic Introduction” Claremont Math Weekend, Claremont, CA	2018
“Perceiving Patterns and Chaos” Invited Colloquium Talk, Colorado College	2017
“Perceiving Patterns and Chaos” Invited Colloquium Talk, Lafayette University	2017
“Perceiving Patterns and Chaos” Invited Colloquium Talk, Agnes Scott College	2017
“Skew Product Flows and Hybrid Systems” Applied Mathematics Seminar, Pomona College, CA	2017
“Skew Product Flows and Hybrid Systems” Women in Math in Southern California Conference, Los Angeles, CA	2017
“Symbolic Dynamics and Deterministic Hybrid Systems” (poster) Joint Mathematics Meetings, San Antonio, TX	2015
“Symbolic Dynamics and Deterministic Hybrid Systems” Mathematics Colloquium, University of Augsburg, Augsburg, Germany	2013

OTHER PRESENTATIONS

“Panel Industry and Research in CSTEM” Associated Students Inc. Invited Panelist	2022
“STEM and Twitter” CSUSM oSTEM Chapter Presentation	2022
“The Math of Climate Science” Climate Teach in, CSUSM, CA	2022
“Queering Math” Mathematics Community Seminar, Department of Mathematics, Pomona College, CA	2018
“Queering Math” FoRCES Reading Group, Pomona College, CA	2017

PROFESSIONAL WORKSHOPS

Harvard Master Class Teaching Math Modeling for Life Science Teaching Math Modeling for Life Science	Summer 2023 Funded Senior Participant
CSUSM Faculty Learning Community Contemplative Pedagogy for Social Justice Contemplative Pedagogy for Social Justice	Fall 2021 Accepted Participant
ICERM Virtual Workshop Advanced Workshop in Data Science for Mathematical Sciences Faculty	June 2021 Accepted Participant
MAA Virtual Workshop How and Why Should Sustainability Be Part of What We Teach?	April 2021 Accepted Participant
Joint Mathematics Meetings, Denver, CO AWM Workshop: Moving Toward Action	January 2020 Funded Participant
Joint Mathematics Meetings, Baltimore, MD MAA Minicourse: Start Teaching Statistics using R and RStudio	January 2019 Funded Participant
Mathematical Association of America MathFest 2018, Denver CO Project NExT Participant	August 2018 Funded Participant
Iowa State University NSF-Sponsored REU Program	Summer 2009 Funded Participant

CSUSM PROFESSIONAL SERVICE

Student Affairs Committee CSTEM Representative	2024-present
STEM Strategic Planning Committee on AI Member	2024-2025
Academic Senate Senator representing CSTEM	2023-2025
San Marcos Informal Mathematics In-person Colloquium (SMIMIC) Co-Coordinator	2023-present
Reid Lecture Committee	2022-present
Math 150 Course Coordinator	2024 - present
Math 132 Community of Practice Tenure-Track Faculty Participant	Fall 2023
Faculty Center Teaching and Learning Group Group member	2022-present
AWM Student Chapter Faculty Advisor	2021 - present
oSTEM Chapter Faculty Advisor	2021 - present
Committee to create a new life sciences calculus Lead Committee Member	Summer 2022
Cal-Bridge Program Cal State Mentor	2022 - 2023
2021-2022 Math Department Search, Tenure Track Line Search Committee Member	2021-2022
Math 132 Course Coordinator	2021-2022

OTHER PROFESSIONAL SERVICE

AWM Executive Committee Social Media Coordinator	2024 - present
MAA SoCal-Nevada Section Vice Section Chair/Section Chair/Past Section Chair	2023-2026
AMS Special Session, JMM 2024 Organizer	2024
SIAM Conference on Applications of Dynamical Systems Mini-Symposium Organizer	2023
AMS Special Session, JMM 2023 Organizer	2023
CSU JMM, CSU Northridge Plenary Talk Chair	2022
Johns Hopkins University Press, Kyne Santos Untitled Book Book Reviewer	2022
CSU JMM Scientific Committee Committee Member	2022
AMS Special Session, JMM 2022 Organizer	2022
AMS Math Reviews Reviewer	2020-present
AWM Media Committee Member	2019-present
Pomona Academy for Youth Success (PAYS) Math Instructor	2017 - 2021
Carroll College Safe Zone Training Co-Facilitator	2021

AWM The First Fifty Years: Reminiscences, Participant History, and Visions for the Future 2020
Article Referee

Carroll College Math Debate 2020
Debater: Who is the most influential mathematician who died young? (Debate Winner Representing Srinivasa Ramanujan)

AMS Social, JMM 2020 2020
Performer, Violinist

JMM MAA Workshop: Identifying and Managing Microaggressions in the Academic Setting 2020
Small group discussion facilitator

EDGE (Enhancing Diversity in Graduate Education) 2019
MATLAB Mini Course Instructor

Project NExT Session on Math Education at 2019 JMM 2019
Facilitator

AWM Strategic Task Force on Diversity and Inclusion 2018
Committee Member

Pomona College 1st Annual Sonia Kovalevsky Day 2018
Organizer

Nebraska Conference for Undergraduate Women in Mathematics 2013
Invited Panelist, Life in Graduate School

Bowdoin College Student Math Representative 2009-2010

MEDIA APPEARANCES

Jeopardy! 2026
Contestant

LGBTTech 2022
PATHS In-Person Networking Event Invited Panelist

My Favorite Theorem 2022
Podcast Guest

LGBTTech 2022
PATHS Program Interview

UNDERGRADUATE STUDENT PROJECTS

Haley Lorenz and Carmen Gutierrez	Summer 2023
“Visualizing the Invariant Distribution of the Stochastic Logistic Map”	
Elisa Zepeda	2020-2021
“Random Forests for Prenatal Maternal Data”	
Eisen Ipac	2018-2019
“Population dynamics in crime dynamics modeling”	
Benjamin Gregory	2017-2018
“Symbolic dynamics of Smale’s horseshoe”	
Abigail Gardner	2016-2017
“Variation of parameters in crime dynamics modeling”	

MASTER’S STUDENT THESIS PROJECTS

Alireza Pakravan	2021-2022
“Solving PDEs Using Neural Networks”	

CSUSM COURSES TAUGHT

Spring 2026:

Math 150 - Calculus for Life Sciences
Math 350 - Foundations for Theoretical Mathematics
Math 530 - Measure Theory

Fall 2025:

Math 350 - Foundations for Theoretical Mathematics
Math 362 - Differential Equations
Math 378 - Number Systems

Spring 2025:

Math 150 - Calculus for Life Sciences (2 Sections)
Math 378 - Number Systems

Fall 2024:

Math 150 - Calculus for Life Sciences

Math 378 - Number Systems

Spring 2024:

Math 350 - Foundations for Theoretical Mathematics

Math 364 - Intermediate Linear Algebra

Fall 2023:

Math 350 - Foundations for Theoretical Mathematics

Math 430 - Real Analysis

Spring 2023:

Math 350 - Foundations for Theoretical Mathematics

Math 430 - Real Analysis

Fall 2022:

Math 350 - Foundations for Theoretical Mathematics

Math 448 - Mathematical Methods in Biology

Spring 2022:

Math 132 - Survey of Calculus

Math 362 - Differential Equations

Fall 2021:

Math 132 - Survey of Calculus

Math 346 - Mathematical Methods of Physics

COURSES TAUGHT (OTHER INSTITUTIONS)

Carroll College:

Math 121 – Differential Calculus (Fall 2019, Fall 2020)

Math 122 – Integral Calculus (Spring 2020, Spring 2021)

Math 207 – Introduction to Statistics (Fall 2019, Spring 2020, Spring 2021)

Math 233 – Multivariable Calculus (Fall 2019, Fall 2020)

Math 306 – Real Analysis (Spring 2020, Spring 2021)

Math 401 – Abstract Algebra and Modern Geometry (Fall 2020)

Pomona College:

Math 30 – Differential Calculus (Spring 2018)

Math 31 – Integral Calculus (2 sections, Fall 2016, Fall 2018)

Math 32 – Multivariate Calculus (2 sections, Spring 2017, Fall 2017)

Math 58 – Introduction to Statistics (Spring 2019)

Math 60 – Linear Algebra (Fall 2017, Spring 2018)

Math 101 – Introduction to Real Analysis (Fall 2018, Spring 2019)

Math 102 – Differential Equations with Modeling (Fall 2016)

PROFESSIONAL MEMBERSHIPS

American Mathematical Society
Mathematical Association of America
Association for Women in Mathematics
National Association of Mathematicians
SACNAS
Spectra

COMPUTER SKILLS

Python
R
 $\text{\LaTeX}$
MATLAB
Mathematica
Git